

CaRS Data Summary — 12 European Electricity Markets

Date Range: 2015-01-01 – 2026-03-09 — Target: Day-ahead electricity price (EUR/MWh)

Per-Country Dataset Summary (Hourly frequency, ~3M total observations)

Code	Country	Observations	Features	Missing %
AT	Austria	312,600	166	<0.01
BE	Belgium	391,296	160	<0.01
CZ	Czech Republic	143,469	153	0.01
DE	Germany	391,686	185	<0.01
DK	Denmark	218,637	168	<0.01
ES	Spain	197,469	154	0.02
FR	France	175,582	161	0.01
HU	Hungary	391,290	152	<0.01
IT	Italy	129,069	155	0.01
NL	Netherlands	392,256	157	<0.01
PL	Poland	143,469	164	0.01
SE	Sweden	109,485	128	0.01
Total		2,996,308	128–185	<0.02

Feature Groups (DE reference, 185 features):

Group	#	Examples
Generation	33	Nuclear, Solar, Wind, Gas
Lag & Rolling	31	Price/load/storage lags
Calendar	30	DoW, holiday, season, hour
Cross-border	24	Flows to/from 8 neighbors
Weather	13	Temp., wind, radiation
Commodities	11	Nat. gas, Brent, WTI
Gas Storage	9	Fill %, inject., withdrawal
Macroeconomic	7	GDP, CPI, unemployment
Price	7	Price, returns, lags
Outages	6	Planned/unplanned (MW)
Oil Fundamentals	5	OPEC basket, inventories
Sentiment	4	VIX, EPU, fear/greed
Trade	2	EU trade bal., USD/EUR
Transport	1	Baltic Dry Index
Hydrogen	1	Grey H ₂ cost
Load	1	Actual demand (MW)
Total	185	

Notes:

- Feature count varies by cross-border neighbors (DE: 8, SE: 4), generation mix, and gas storage presence
- Missing values handled via forward-fill + backward-fill; clean datasets: 0% missing

Data Sources:

Source	Freq.	Data
ENTSO-E ¹	Hourly	Prices, gen., load, flows
Open-Meteo ²	Hourly	Weather variables
AGSI+/GIE ³	Daily	Gas storage levels
Yahoo Finance ⁴	Daily	Commodities, VIX, BDI
FRED ⁵	Daily	Macro, oil, trade, sent.
Eurostat ⁶	Monthly	EU trade balance
OPEC ⁷	Daily	OPEC basket price
PolicyUncert. ⁸	Daily	Econ. Policy Uncert. Idx

¹transparency.entsoe.eu ²open-meteo.com
³agsi.gie.eu ⁴[yfinance package](#)
⁵fred.stlouisfed.org ⁶ec.europa.eu/eurostat
⁷opec.org ⁸policyuncertainty.com

Price Statistics (EUR/MWh):

Country	Mean	Std	Min	Max
AT	95.3	88.7	−500	920
BE	77.8	79.5	−500	871
CZ	84.8	74.2	−225	871
DE	73.5	80.4	−500	936
DK	93.3	93.2	−223	936
ES	75.7	53.9	−15	700
FR	72.5	72.1	−135	2988
HU	88.8	89.2	−500	1047
IT	103.5	80.0	0	871
NL	77.0	78.3	−500	873
PL	112.2	74.2	−133	1036
SE	45.0	43.3	−60	590

Data Frequencies:

- **Hourly:** Prices, generation, load, flows, weather
- **Daily:** Commodities, gas storage, outages, macro, sentiment, oil, transport, trade
- **Monthly:** EU trade balance (Eurostat)
- Daily/monthly → hourly via forward-fill